# Olivier Juan

Research Engineer Expert

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## Experience

**EDF R&D · OSIRIS, Palaiseau, France** — Jan 2008 – present, 18 years 4 months

- Led the ground-up redesign and production deployment of **Apogène**, EDF's short-term unit commitment software (MILP, branch-and-bound, ADMM, RCSPP, Min Cost Flow). 4 versions deployed in production (V2 2018, V3 2020, V4 2021, V5 2025), reducing overall plant operating costs by several percent.
- Managed a team of 10–25 engineers and researchers across the full lifecycle: mathematical modeling, algorithm design, C++ implementation, CI/CD (GitLab), HPC deployment (Slurm, Singularity).
- Integrated ML-based branching heuristics (RL, GNN) into the solver loop; results published at NeurIPS 2025 and AAAI 2026.
- Developed optimization models for smart EV charging (DREEV joint venture, EDF × Nuvve). DREEV became first French EV aggregator accredited for FCR by RTE (2022).
- Earlier work (2008–2012): task scheduling for maintenance planning, LNG terminal planning & scheduling, combustion and emission optimization for coal power plants.

**École Centrale de Paris, France** — Jan 2007 – Jan 2008, 1 year 1 month

- Combinatorial algorithms for medical imaging, with Nikos Paragios.

**University of Western Ontario, Canada** — Jan 2006 – Jan 2007, 1 year 1 month

- Graph cuts and combinatorial algorithms for computer vision, with Yuri Boykov.

## Education

**École Nationale des Ponts et Chaussées (ENPC), France**, Computer Vision — Jan 2006

- Distinction. Dissertation: *On Some Extensions of Level Sets and Graph Cuts & Their Applications to Image and Video Segmentation*

**ENS Paris-Saclay (formerly ENS de Cachan), France**, Computer Vision — Jan 2002

## Awards

**IndusRO'2022 Prize** — Jan 2022

Awarded for the industrial impact of Apogène on EDF's power plant operations.
ROADEF (French Society for Operations Research and Decision Aid)

## Publications

**PlanB&B: Planning in Branch-and-Bound via Model-Based RL** — Jan 2026

Strang, Alès, Bissuel, Juan, Kedad-Sidhoum, Rachelson.
Paul Strang, Zacharie Alès, Côme Bissuel, Olivier Juan, Safia Kedad-Sidhoum, Emmanuel Rachelson
arxiv.org/abs/2511.09219

**BBMDP: A Markov Decision Process for Variable Selection in B&B** Jan 2025
Strang, Alès, Bissuel, Juan, Kedad-Sidhoum, Rachelson.
Paul Strang, Zacharie Alès, Côme Bissuel, Olivier Juan, Safia Kedad-Sidhoum, Emmanuel Rachelson
arxiv.org/abs/2510.19348

**Reinforcement Learning for Variable Selection in a Branch and Bound Algorithm** Jan 2020
Etheve, Alès, Bissuel, Juan, Kedad-Sidhoum.
Marc Etheve, Zacharie Alès, Côme Bissuel, Olivier Juan, Safia Kedad-Sidhoum
doi.org/10.1007/978-3-030-58942-4_12

## Skills

**Optimization:** MILP, Branch-and-bound, Cutting planes, RCSPP, ADMM, Min Cost Flow, Constraint programming, Decomposition methods, Heuristics, Genetic Algorithm

**Solvers:** CPLEX, GUROBI, SCIP, FICO Xpress, COIN-OR, HiGHS, Interior Point Methods

**Machine Learning:** Reinforcement learning, Graph Neural Networks, ML for combinatorial optimization

**Frameworks:** PyTorch, PyG, RLLib, Ray

**Languages:** C++, Python, TBB, Julia

**Engineering:** GitLab CI/CD, Docker, HPC / Slurm, Singularity, CMake, Conan